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Surveillance of live birds for active and past infections reveals the impact of highly pathogenic H5N1 on seabird populations in Atlantic Canada

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Abstract

Highly pathogenic avian influenza clade 2.3.4.4b H5Nx viruses (HPAIVs) have caused significant mortality in wild birds. We investigated active avian influenza virus (AIV) infections and assessed seroprevalence based on general nucleoprotein-specific and H5-specific antibodies in seabirds in Atlantic Canada during 2022–2023. Results were compared to seroprevalence data from 2011 to 2016 to evaluate changes in infection rates following the incursion of HPAIV. We sampled 1073 wild live birds belonging to 14 species. Active H5Nx infections were detected only in 2022, with the highest prevalence in Common Murres (64%), followed by Black-legged Kittiwakes (13%). Differences in seroprevalence were observed between years and species, with both Common Murres and Atlantic Puffins exhibiting higher rates after HPAIV incursion, particularly for anti-H5-specific antibodies. Seroprevalence also differed between locations, and age-related differences in exposure were observed with juvenile birds having much lower seroprevalence than adult birds, particularly among gulls. Despite no active infections being detected in 2023, high seroprevalence across species suggests persisting immunity and/or recent virus circulation. These findings underscore the importance of serological monitoring in tracking AIV dynamics, as antibody detection provides critical insights into past HPAIV exposure even when active infection rates are low.

Key words: avian influenza virus, surveillance, serology, seabirds, gulls, immunity

Introduction

Influenza A viruses (IAVs) circulate regularly in wild avian reservoir hosts, including waterfowl, shorebirds, gulls, and other seabirds (Hinshaw et al. 1982; Olsen et al. 2006; Lang et al. 2016). The emergence of highly pathogenic avian influenza virus (HPAIV) of the A/goose/Guangdong/1/1996 (H5N1) lineage (Guo et al. 1998) led to a new reality for global avian influenza virus (AIV) dynamics (Ramey et al. 2022; Xie et al. 2023). Between 1997 and 2003, the virus spread extensively among poultry in Asia (Chan 2002; Guan et al. 2002), and in 2005 caused an outbreak among wild birds in China (Chen et al. 2005), spreading to wild birds in other regions of Asia, Europe, and Africa in subsequent years (Sims et al. 2005; Olsen et al. 2006; Salzberg et al. 2007). During this time, the virus repeatedly infected wild birds, although it was not maintained in these populations (Wille and Waldenström 2023). Its continued circulation and emergence led to the evolution of sub-lineage 2.3.4.4 (Smith et al. 2015) and then sub-lineage 2.3.4.4b (The Global Consortium for H5N8 and Related Influenza Viruses 2016) in the early 2010s. Generally, infections would cause large-scale mortalities in the wild bird populations and result in finite outbreaks. However, clade 2.3.4.4

viruses were highly adapted to wild birds and viral circulation in these populations became persistent. Circulation of clade 2.3.4.4b H5Nx viruses in wild birds in Europe continued through 2020 and into 2021 (Lewis et al. 2021; Pohlmann et al. 2022), which was then followed by the dispersion of H5N1 HPAIVs from Eurasia into North America (Alkie et al. 2022; Caliendo et al. 2022*b*).

Following the Atlantic introduction in late 2021, the virus rapidly spread across North America (Bevins et al. 2022; Youk et al. 2023; Giacinti et al. 2024; Jahid and Nolting 2025). Multiple subsequent incursions of HPAIV into North America, including H5N1 and H5N5 strains, from Eurasia have occurred through both the Atlantic and Pacific routes (Alkie et al. 2022; Erdelyan et al. 2024; Rahman et al. 2024; Ramey et al. 2024). In late 2022, the virus reached South America, causing widespread outbreaks (Leguia et al. 2023; Ruiz-Saenz et al. 2023; Bruno et al. 2025), ultimately arriving in Antarctica (Banyard et al. 2024). To date, Australia is the only geographic region where the virus has not been detected (Plaza et al. 2024, 2025).

The virus diversified greatly during its spread (Kandeil et al. 2023; Signore et al. 2025). The ancestral H5N1 (genotype A1) virus (Signore et al. 2025) and its derived reassortants have resulted in numerous mass mortality events in wild birds of various species across the Americas (Gamarra-Toledo et al. 2023; Ahlstrom et al. 2024; Avery-Gomm et al. 2024; Sacristán et al. 2024) and has been transmitted to a wide diversity of terrestrial and marine mammals (Bordes et al. 2023; Leguia et al. 2023; Caserta et al. 2024).

A wide diversity of seabird species has been significantly impacted by the increased geographic spread of HPAIVs (Banyard et al. 2022; Wille and Waldenström 2023; Caliendo et al. 2024). Seabirds represent long-lived, often colonial nesting species that spend the majority of their lives in marine environments and return to the land primarily for breeding (Gaston 2004). Their wide distribution, high mobility, and colonial breeding play important roles in shaping AIV ecology among seabirds (Lang et al. 2016). In Eastern Canada, Common Eiders (*Somateria mollissima*), Northern Gannets (*Morus bassanus*), Common Murres (*Uria aalge*), Atlantic Puffins (*Fratercula arctica*), Black-legged Kittiwakes (*Rissa tridactyla*), and gulls (mostly Herring Gulls (*Larus argentatus*)) were some of the species with the highest observed mortalities during outbreaks in 2022 (Taylor et al. 2023; Avery-Gomm et al. 2024).

Following extensive circulation among wild birds, HPAIVs of clade 2.3.4.4b appear to have become enzootic in Asia and Europe and are frequently detected among seabirds (Pohlmann et al. 2022; Caliendo et al. 2024; Lean et al. 2024). Multiple incursions of HPAIVs to North America via Atlantic Canada (Rahman et al. 2024) makes eastern Canada a region of key importance for AIV surveillance. Furthermore, eastern Canada hosts a significant number of seabirds that breed across the North Atlantic Ocean (Hatch et al. 2020; Lowther et al. 2020; Ainley et al. 2021; d'Entremont et al. 2022) and hence the spread of highly pathogenic avian influenza (HPAI) among these species has larger ecological implications. For instance, Atlantic Canada has the biggest Leach's Storm Petrel (*Hydrobates leucorhous*) colony (Pollet et al. 2021) and the second biggest Atlantic Puffin colony (Lowther et al. 2020) in the world.

In this study, we performed active surveillance on live birds for current AIV infections and used serology to determine past infection status. Serological studies offer a valuable insight into exposure to pathogens within wildlife populations as antibodies can persist long after initial infection (Jourdain et al. 2010; Gilbert et al. 2013; Verhagen et al. 2015), providing information about previous pathogen exposure that cannot be captured through surveillance for active infections alone (Kim et al. 2014; Wight et al. 2024). Sera from birds sampled following the incursion of HPAIV into Atlantic Canada and archived samples from before the incursion were tested for antibodies against AIV broadly (anti-nucleoprotein, NP) and specifically against haemagglutinin (HA) subtype H5. Using these data from a broad diversity of seabird species sampled from different populations allowed for examination of how AIV exposure among these species varied temporally, geographically, and between age groups and how it has changed following the arrival of HPAIV into North America.

Methods

Ethics statement

Bird capture and sampling was performed under Federal Bird Banding and Scientific Research Permits 10332, 10559, 10926, and 10851 following guidelines of the Canadian Council on Animal Care with approved protocols 11-01-AL, 12-01-AL, 13-01-AL, 14-01-AL, 17-05-AL, 20-05-AL, and 22-01-PB from Memorial University's Institutional Animal Care Committee and 15GR01, 16GR01, 22GR01, 22RR01, 23GR01, 23JR02, and 23RR01 from Environment and Climate Change Canada and the Wildlife Eastern Animal Care Committee. Sampling in Nunatsiavut, Labrador was conducted under Nunatsiavut Government Research Advisory Committee (NGRAC) permits NGRAC-20117007 and 18824526. Laboratory work at Memorial University of Newfoundland was performed under biosafety permit S-103 from Memorial University's Institutional Biosafety Committee. Further serological testing was conducted at the National Centre for Foreign Animal Disease (NCFAD).

Sampled species

Birds were live-captured using approved protocols which varied depending on the target species and banded with a metal leg band issued by the Canadian Wildlife Service Bird Banding Office prior to release. Age group was determined for all birds using plumage aspect (Pyle 2008; Pyle et al. 2022). Individuals were grouped as juveniles if they were aged hatch year or local, and all other individuals were grouped into another category as sub-adults and adults. All sampling occurred between January and September in Atlantic Canada, with most sampling occurring primarily during the breeding season (between May and September). Sampling occurred from 2011 to 2016 and 2022 to 2023.

Larus gull species are common vectors of AIV (Kawaoka et al. 1988; Arnal et al. 2015); therefore, we grouped all *Larus* gulls species together, hereafter referred to as "gulls", while

all non-*Larus* seabirds are hereafter referred to as "seabirds". Sampled seabird species included Atlantic Puffins, Arctic Terns (*Sterna paradisaea*), Black Guillemots (*Cepphus grille*), Black-legged Kittiwakes, Common Murres, Great Shearwaters (*Ardenna gravis*), Leach's Storm Petrels, Northern Gannets, Razorbills (*Alca torda*), and Thick-billed Murres (*Uria lomvia*). Sampled gull species included Glaucous Gulls (*Larus hyperboreus*), Great Black-backed Gulls (*Larus marinus*), Herring Gulls (*Larus argentatus*), and Iceland Gulls (*Larus glaucoides*). Although Black-legged Kittiwakes and Arctic Terns are members of the family Laridae, we grouped them as seabirds in this study due to their highly pelagic nature (Frederiksen et al. 2012).

In total, samples from 916 seabirds and 157 gulls were tested and included in this study (Table S1 and Supplementary Data). Sampling took place across several regions within Atlantic Canada, with most samples from the province of Newfoundland and Labrador, followed by Nova Scotia and New Brunswick (Fig. 1).

Sample collection and processing

For the detection of active infections, paired oropharyngeal and cloacal swab samples were collected from each bird and pooled into a single tube of Multitrans viral transport medium (Starplex Scientific, Product #S160-100) or UTM® Universal Transport Medium™ (Copan, Product #3C047N) and kept cool prior to being frozen at −80 °C. Samples for the detection of active infections were only collected in 2022 and 2023, after the arrival of HPAIV in North America in late 2021.

Samples for serological analysis were collected both prior to and after the arrival of HPAIV in late 2021 for Atlantic Puffins, Black-legged Kittiwakes, Common Murres, Razorbills, and gulls, and only after the arrival of HPAIV for Black Guillemots, Leach's Storm Petrels, and Northern Gannets. Samples were collected prior to the arrival of HPAIV in the years 2011–2016 and samples were collected after HPAIV arrival in the years 2022–2023. Paired swab and blood samples were only available for birds sampled post-2021. For most individual puffins, kittiwakes, murres, gannets, razorbills, and gulls, up to 3 mL of blood was collected from the brachial (wing) or tarsal vein. Serum was separated from clotted blood by centrifugation at 3000 × *g* for 5 min within 2–3 h of sampling or after storing at 4 °C overnight and was stored at −20 °C until analysis.

As an alternative to these larger-volume serum samples, Nobuto filter strips (Advantec MFS, Inc., USA, Product #49010010) were used to collect samples for serological analyses from Black Guillemot, Leach's Storm Petrel, and Razorbill. Giacinti et al. (2025) showed that the results from the two sample types are comparable with optimized thresholds. This allowed for sampling of smaller species for which sufficient blood could not be collected, and in field situations without access to a centrifuge. For filter strip samples, the narrow end of the strip was saturated with fresh whole blood collected from the brachial vein using a needle and capillary tube, left to dry, and stored at −20 °C until processing. Approximately, 0.5 cm pieces were cut from each strip and submerged into 400 μL of phosphate-buffered saline (pH 7.2; ThermoFisher Scientific, USA, Product #20012027). The immersed strips were incubated at 4 °C for approximately 24 h, vortexed and the eluate removed and stored at −20 °C until analysis.

Screening for active AIV infections

RNA was extracted from swab samples using the Qiagen Viral RNA Mini Kit (Qiagen, Product #52906) as per manufacturer's instructions. Real-time reverse transcriptase-polymerase chain reaction (RT-PCR) was used to test for the presence of the IAV matrix gene, and positive samples were subsequently tested for the H5 subtype of the HA gene as previously described by our research group (Wight et al. 2024). In brief, real-time RT-PCR was performed using AgPath-ID™ One-Step RT-PCR reagents (Applied Biosystems, Product #4387424) on either a StepOnePlus or a QuantStudio3 Real-Time PCR System (Applied Biosystems). The primers, probes, and setup and cycling conditions are described in detail in our previous work (Wight et al. 2024). Samples displaying the characteristic amplification curve with a Ct value $\leq$ 45 were considered positive, while those that had the characteristic curve but did not cross the threshold were classified as inconclusive and reported as Ct $>$ 45. All samples that tested positive for H5 RNA were confirmed as the HPAI H5 clade 2.3.4.4b by genomic sequencing at the NCFAD, Canadian Food Inspection Agency.

Serum antibody screening

Antibodies against AIV NP were detected by enzyme-linked immunosorbent assays (ELISAs) using the IDEXX AI MultiS Screen Ab test (IDEXX Canada, Product #99-12119) as per manufacturer's instructions. Serum samples were interpreted as positive for anti-NP antibodies if the sample to negative control ratio (S/N) was <0.5. For Nobuto filter strip samples, the eluate was directly used for the ELISA without further dilution and then treated the same as the serum samples. Filter strip samples were interpreted as positive for anti-NP antibodies if the S/N was <0.7708 (Giacinti et al. 2025). Positive samples were subsequently tested for antibodies against all clades of the H5 subtype, including low pathogenic AIVs (LPAIVs), by competitive ELISA (Hochman et al. 2023) at the NCFAD, Canadian Food Inspection Agency. As such, this ELISA does not discriminate between infection with LPAI and HPAI H5 viruses. A percentage inhibition (PI) $\geq$ 35% was considered positive for serum samples, while a PI $\geq$ 20.37% was considered positive for filter strip samples (Giacinti et al. 2025).

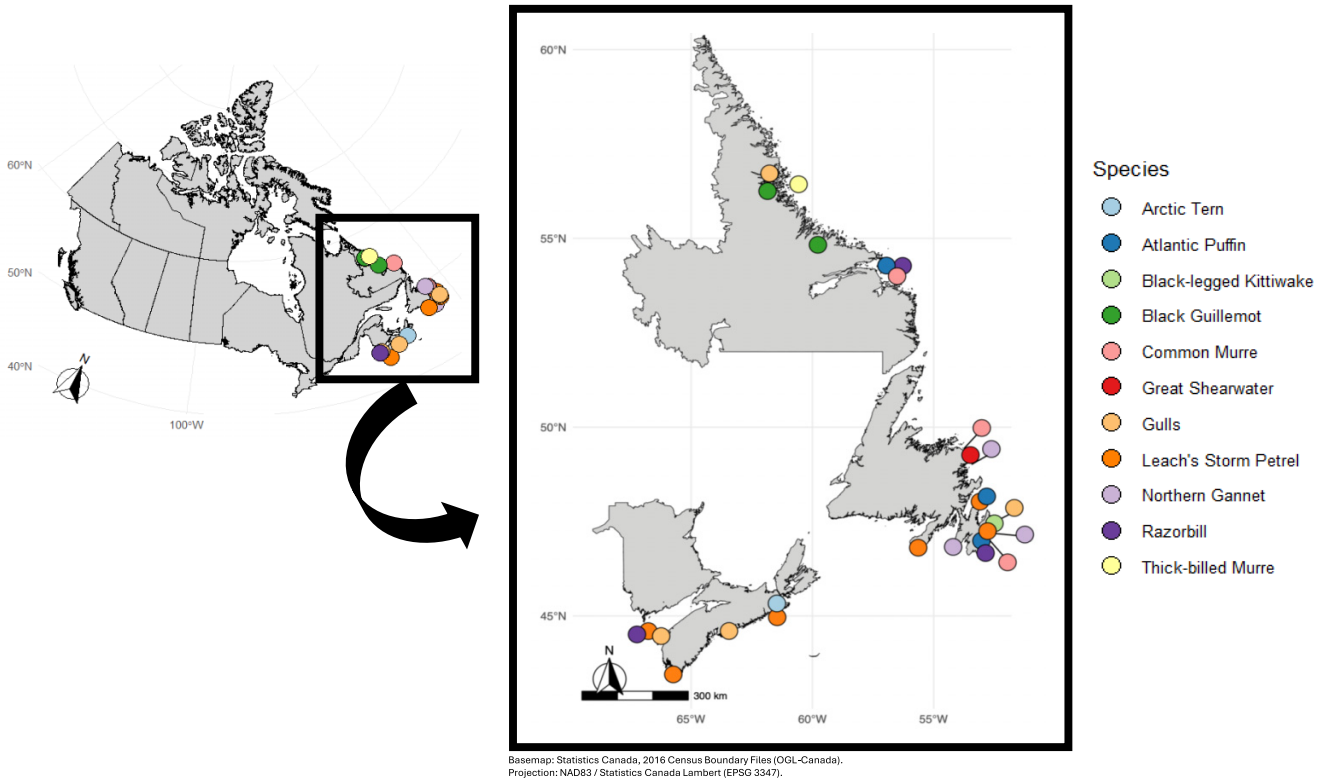
We note that Northern Gannet blood samples collected in 2023 were mistakenly diluted with viral transport medium. Testing viral transport medium alone by ELISA yielded high S/N ratios (>1), suggesting contribution from the medium to the positivity of the results is negligible.

Data analysis and visualization

Data were analyzed and visualized using R v4.4.0 (R Core Team 2024) with the packages ggplot2 v3.5.1 (Wickham et al. 2007), sf v1.01.16 (Pebesma 2016), and dplyr v1.1.4 (Wickham et al. 2007).

Fig. 1. Distribution of sampling sites and birds sampled for avian influenza virus surveillance. Map of the sampling locations and species (A) and table displaying the number of each species that was sampled at each location (B).

A



B

Region	Location	Arctic Tern	Atlantic Puffin	Black Guillemot	Black-legged Kittiwake	Common Murre	Great Shearwater	Gulls	Leach's Storm Petrel	Northern Gannet	Razorbill	Thick-billed Murre
Labrador	Nain			7				22				
	Pyramid Islands											25
	Postville			16								
	Gannet Islands		25			10					15	
Northeast Coast of Newfoundland	Musgrave Harbour									1		
	Deadman's Bay					1				1		
	Cape Freels						1					
	Baccalieu Island		18						92			
	Bay de Verde								9			
St. John's area	White Hills							5				
	Quidi Vidi							23				
	St. John's harbour southside							15				
East Coast of Avalon Peninsula	Maddox Cove									1		
	Petty Harbour							2				
	Witless Bay							18				
	Gull Island		129		91	149		43	56		8	
Southwest Coast of Newfoundland	Cape St. Mary's									26		
	Middle Lawn Island								95			
Nova Scotia	Country Island	10						15	13			
	Devil's Island							10				
	Bon Portage Island								20			
New Brunswick	Kent Island								26			
	Machias Seal Island										10	

Results

Active infections

Active AIV infections were detected only in 2022 for Black-legged Kittiwakes, Atlantic Puffins, Northern Gannets, and Common Murres; no active infections were detected in any birds sampled in 2023 (Table 1). All AIV matrix-positive samples from 2022 were identified as the H5 subtype by real time RT-PCR. Among the species that tested positive in 2022, Common Murres exhibited the highest prevalence rate, with 64% (27/42) positive for AIV, followed by Black-legged Kittiwakes at 13% (5/39) (Table 1). For both Atlantic Puffins and Northern

Table 1. Prevalence of active avian influenza virus infection in seabirds and gulls after highly pathogenic avian influenza virus arrival.

Species	Year	Number of samples	Number positive (percentage %)	
			Matrix RNA	H5 RNA
Arctic Tern	2023	10	0	0
Atlantic Puffin	2022	64	1 (2%)	1 (2%)
	2023	58	0	0
Black Guillemot	2023	23	0	0
Black-legged Kittiwake	2022	39	5 (13%)	5 (13%)
	2023	10	0	0
Common Murre	2022	42	27 (64%)	27 (64%)
	2023	29	0	0
Great Shearwater	2022	1	0	0
Gulls (juveniles)	2022	4	0	0
	2023	27	0	0
Gulls (sub-adults and adults)	2022	2	0	0
	2023	44	0	0
Leach's Storm Petrel	2022	173	0	0
	2023	97	0	0
Northern Gannet	2022	18	1 (5%)	1 (5%)
	2023	11	0	0
Thick-billed Murre	2023	25	0	0

Gannets, only one individual from each species tested positive for AIV in 2022 (Table 1).

Overview of seroprevalence

There was variability in seroprevalence across years for most species (Figs. 2 and 3), with Atlantic Puffins and Common Murres exhibiting higher seropositivity each year for anti-NP antibodies following the incursion of 2.3.4.4b H5Nx virus compared to the years before the arrival of HPAIV in 2021 (Fig. 2). For Common Murres, H5-specific antibodies were only detected after HPAIV arrival between 2022 and 2023 (Fig. 2C), while for Atlantic Puffins, Black-legged Kittiwakes, Razorbills, sub-adult and adult gulls, H5-specific antibodies were detected between 2013 and 2015, although at low levels (Figs. 2 and 3). Higher seropositivity was observed for sub-adult and adult gulls in comparison to juvenile gulls across all years (Fig. 3).

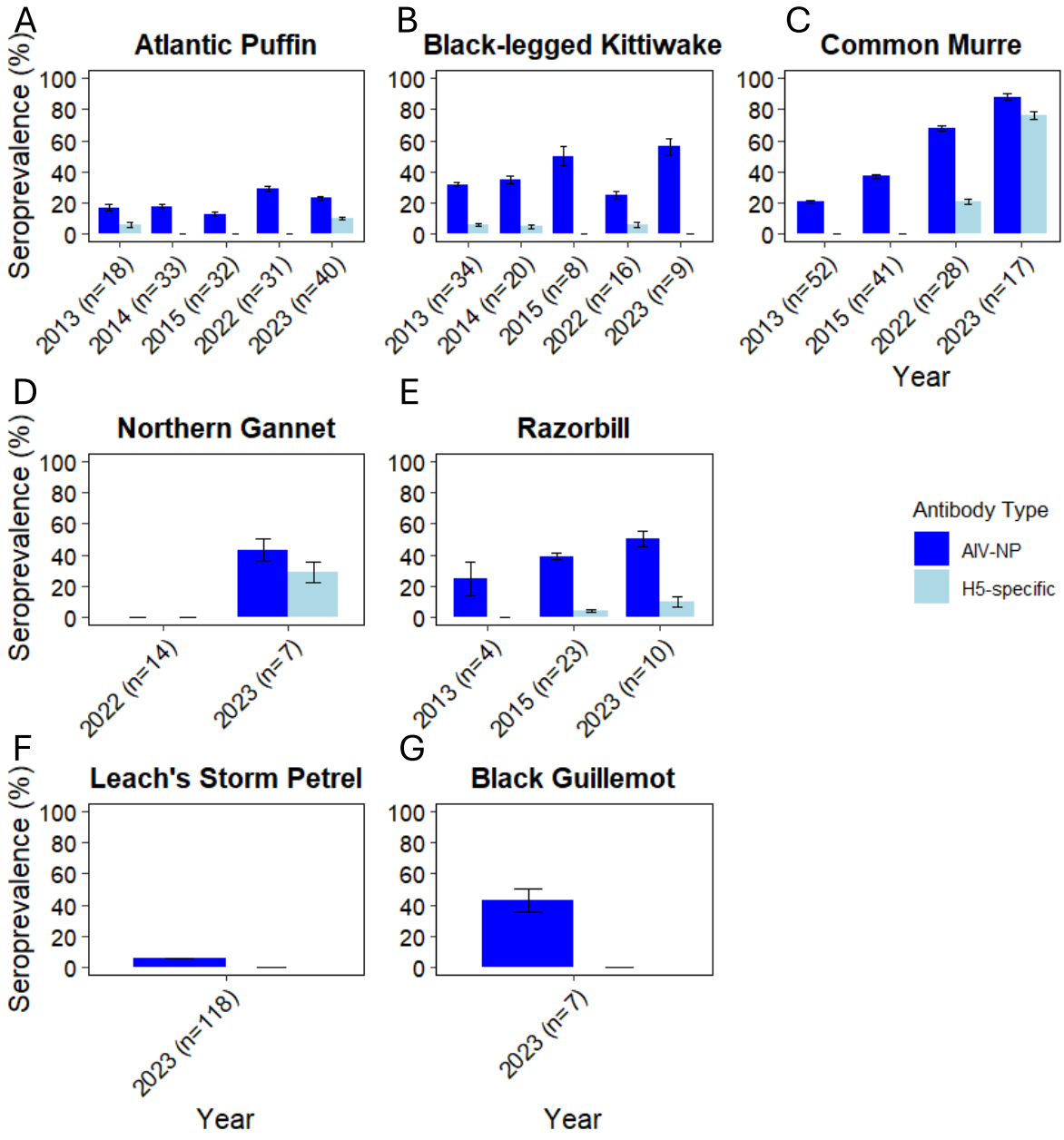
We also looked at seroprevalence by age (Fig. S1) and location (Figs. S2 and S3) for certain species where it was possible to perform these comparisons. General AIV seroprevalence was lower in juvenile birds compared to sub-adults and adults for seabirds (Fig. S1) and gulls (Fig. 3). Furthermore, no anti-H5 antibodies were detected in juveniles. While there were some differences in seroprevalence between locations within the same species, the rates at different locations were similar for the same species within the same year (Figs. S2 and S3).

AIV prevalence by species

Common Murre

Before the incursions of HPAIV into Atlantic Canada, anti-NP seroprevalence among Common Murres was 21% in 2013 at Gull Island and 37% in 2015 at the Gannet Islands, while all murres were anti-H5 seronegative in both years (Fig. 2C). In 2022, the majority of the swab samples from Common Murre collected at Gull Island in late July tested positive for H5 RNA (25/31), indicating an active outbreak at that time. Blood samples were available for 16 of the 25 birds that were actively infected. Fifty percent (8/16) of these actively infected birds tested positive for anti-NP antibodies and two of those were also positive for anti-H5 antibodies (see Supplementary Data). Roughly 2 weeks later, in early August, all sampled Common Murres ($n = 10$) from Gull Island tested positive for anti-NP antibodies, with four individuals (4/10) also positive for anti-H5 antibodies. Only 1 of those 10 birds from early August was also actively infected at the time of sampling but negative for anti-H5 antibodies. Elsewhere, a single Common Murre sampled at Deadman's Bay in early August tested positive for H5 RNA, suggesting that the virus was circulating concurrently at multiple colonies around Newfoundland in 2022. We did not have a blood sample from the bird in Deadman's Bay. The overall anti-NP and anti-H5 seroprevalence in 2022 was 68% and 21%, respectively (Fig. 2). In 2023 at Gull Island, anti-NP seroprevalence further increased to 88% and anti-H5

Fig. 2. Seroprevalence of anti-NP and anti-H5 antibodies in seabirds before (2013–2015) and after (2022–2023) highly pathogenic avian influenza virus arrival in Atlantic Canada. Data are presented as the mean seroprevalence ± standard error. Both AIV-NP and H5-specific seroprevalence are shown relative to all samples tested per year. AIV, avian influenza virus; NP, nucleoprotein.

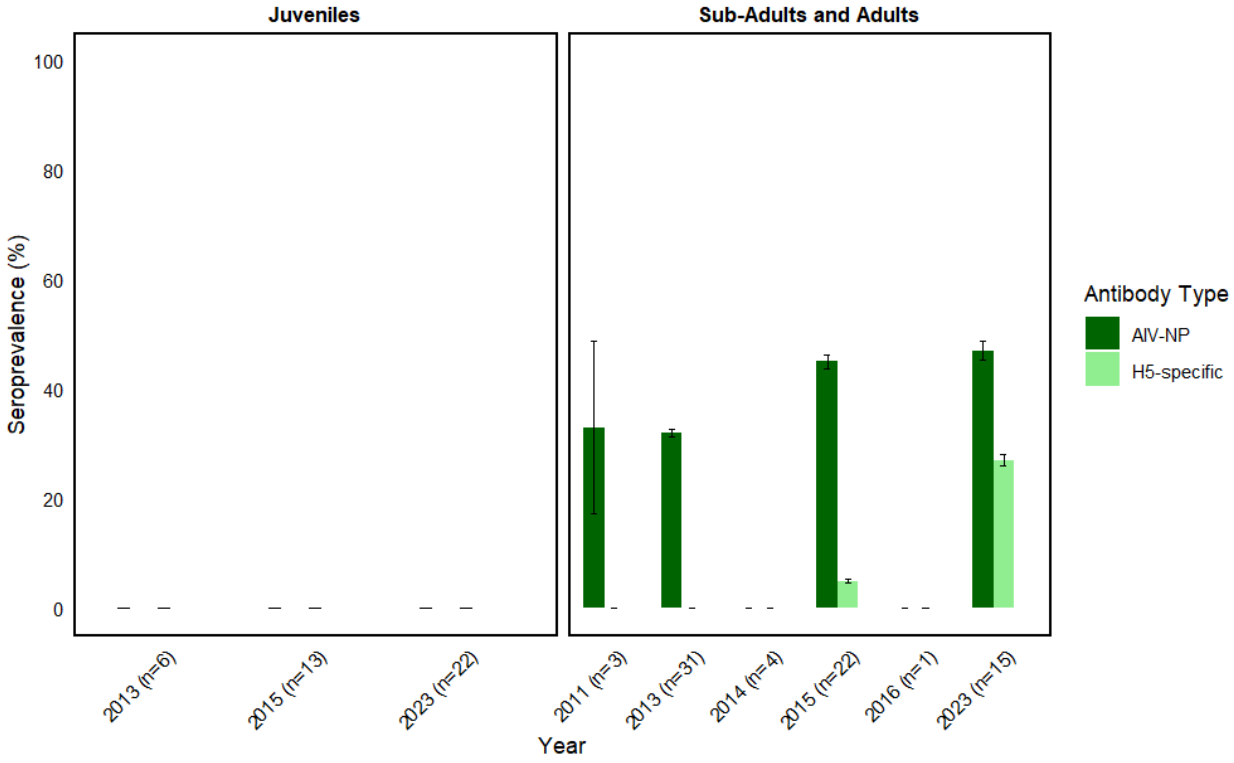


seroprevalence to 76%; however, no active infections were detected during this period (Fig. 2C; Table 1).

Black-legged Kittiwake

Before the arrival and spread of HPAIV, anti-NP seroprevalence in Black-legged Kittiwakes at Gull Island in 2013, 2014, and 2015 was 32%, 35%, and 50%, respectively, with anti-H5 seroprevalence below 10% in each of these years (Fig. 2B). This suggests that Black-legged Kittiwakes are regularly infected with LPAIVs at Gull Island, including with low pathogenic H5 viruses. In 2022, no active infections were detected in late June. However, by late July, five Black-legged Kittiwakes tested positive for H5 RNA, aligning with active infections detected in Common Murres at the colony and indicating the arrival of HPAI (H5N1, genotype A1) (Signore et al. 2025). Among those five birds, three had corresponding blood samples and all were negative for anti-H5 antibodies, but one was positive for anti-NP antibodies (see Supplementary Data). While seroprevalence rates in 2022 (25% for anti-NP antibodies and 6% for anti-H5 antibodies) were comparable to rates pre-HPAIV arrival (Fig. 2B), this was likely affected by sampling occurring at the same time as ongoing active infections when individuals had not yet seroconverted. At the same colony in

Fig. 3. Seroprevalence of anti-NP and anti-H5 antibodies in gulls before (2011–2016) and after (2023) highly pathogenic avian influenza virus arrival in Atlantic Canada. Data are presented as the mean seroprevalence ± standard error. Both AIV-NP and H5-specific seroprevalence are shown relative to all samples tested per year. AIV, avian influenza virus; NP, nucleoprotein.



July 2023, anti-NP seroprevalence among Black-legged Kittiwakes had increased to 56% and all birds were seronegative for anti-H5 antibodies.

Atlantic Puffin

Before the arrival of HPAIV, anti-NP seroprevalence in Atlantic Puffin was 17% in 2013 and 18% in 2014 at Gull Island, and 13% in 2015 at the Gannet Islands, while the seroprevalence of anti-H5 antibodies was 6% at Gull Island in 2013 (Fig. 2A). Therefore, anti-NP seroprevalence was roughly the same in the years before the arrival of HPAIV. In 2022, while no active infections were detected at Gull Island in June, one Atlantic Puffin sampled in mid-July tested positive for H5 RNA. We did not have a blood sample from this puffin, so it is unknown if the bird also had anti-IAV antibodies. From mid- to late-July of 2022, anti-NP antibodies were detected in Atlantic Puffins at Gull Island and Baccalieu Island (Fig. S2B). The overall anti-NP seroprevalence was 29% in 2022, slightly higher than the rates observed pre-HPAIV arrival in the region (Fig. 2). Anti-H5 antibodies were not detected in Atlantic Puffins in 2022, likely reflecting the timing of sampling as also suspected for Black-legged Kittiwakes. In 2023, both anti-NP and anti-H5 antibodies were detected at Gull Island over a period of 6 weeks from mid-June to late-July. The overall seroprevalence for 2023 was 23% for anti-NP and 10% for anti-H5 antibodies (Fig. 2A).

Northern Gannet

No samples from Northern Gannet were available prior to the arrival of HPAIV in the region. Neither anti-NP nor anti-H5 antibodies were detected in Northern Gannets in 2022 (0/14; Fig. 2D). Birds sampled in early July 2022 at Cape St. Mary's tested negative for active infection and anti-NP antibodies, perhaps due to most of these being juvenile birds. Furthermore, according to Careen et al. (2024), the outbreak at that colony began on approximately July 17, after samples for this study were collected. In 2023, anti-NP seroprevalence was 43% and anti-H5 seroprevalence was 29% among Northern Gannets at Cape St. Mary's, though this seroprevalence may be underestimated due to these samples being mistakenly diluted (see the "Methods" section).

Razorbill

Before the arrival of HPAIV, the seroprevalence among Razorbills was 25% and 0% in 2013 at Gull Island, and 39% and 4% in 2015 at Gannet Islands, of anti-NP and anti-H5 antibodies, respectively (Fig. 2E; Fig. S2C). At Machias Seal Island in New Brunswick in 2023, anti-NP and anti-H5 seroprevalence among Razorbills was 50% and 10%, respectively (Fig. 2E). Importantly, the three Razorbill colonies that were sampled are geographically distant from each other, limiting direct comparisons (Fig. 1).

Leach's Storm Petrel

No samples from Leach's Storm Petrels were available prior to the arrival of HPAIV in the region. Swab samples were collected in both 2022 and 2023 and blood samples were collected between June and September 2023 at Middle Lawn Island and Baccalieu Island on the island of Newfoundland, Country Island and Bon Portage Island in Nova Scotia and Kent Island in New Brunswick. No active infections were detected in either 2022 or 2023 (Table 1). Filter strip blood samples were collected from 118 birds in 2023 and subjected to serological analyses, of which 7 were positive for anti-NP antibodies (6%) but negative for anti-H5 antibodies (Fig. 2F) using the optimized cut-off for use with filter strip samples (Giacinti et al. 2025).

Black Guillemot

No samples from Black Guillemots were available prior to the arrival of HPAIV in the region. Among seven birds sampled in Nunatsiavut in July 2023, anti-NP and anti-H5 seroprevalence was 43% and 0%, respectively (Fig. 2G). The absence of detectable anti-H5 antibodies likely indicates that birds were exposed to LPAIV that were non-H5 viruses. However, given that anti-H5 antibodies have been shown to wane more quickly than anti-NP antibodies (Wight et al. 2024), these birds may instead have not been recently exposed to H5 viruses.

Gulls

Among sub-adults and adults, anti-NP seroprevalence was between 33% and 47% for most years (2011, 2013, 2015). However, all birds sampled in 2014 and 2016 were seronegative (Fig. 3). Anti-H5 seroprevalence was highest in 2023 at 27%, following the arrival of HPAIV. Prior to 2022, H5 antibodies were only detected in 2015, with a low seroprevalence of 5%.

No active infections were detected in 2022 or 2023 (Table 1). As no serum was collected from gulls in 2022, the seroprevalence of anti-NP antibodies in 2022 is unknown. In 2023, swab samples were collected from Herring Gulls at Country Island in late May and at Gull Island between late May and early June, and paired swab and serum samples were collected from Great Black-backed, Glaucous, and Herring Gulls breeding on islands around Nain, Nunatsiavut between late June and mid-July. Approximately half of these gulls (7/13) were seropositive for anti-NP antibodies, of which four also had detectable anti-H5 antibodies. The overall anti-NP and anti-H5 seroprevalence for sub-adult and adult gulls in 2023 was 47% and 27%, respectively (Fig. 3). Despite no detections of active infections in 2023, the presence of anti-H5 antibodies indicates gulls sampled in Nain had been infected sometime prior to when sampling occurred.

When comparing seroprevalence rates between gull species, both anti-NP and anti-H5 seroprevalence were highest in Great Black-backed Gulls (see Supplementary Data), although this may be due to a small sample size ($n = 6$ between 2011, 2013, and 2023). The lowest anti-NP seroprevalence was observed in Iceland Gulls ($n = 14$ between 2013 and 2015). Among Herring Gulls, anti-NP antibodies were detected in all years except for 2011 and 2016, although only a single bird was sampled in each of these years. Only Herring Gulls had detectable anti-H5 antibodies prior to the introduction of HPAIV, presumably due to previous infections with LPAI H5 viruses. In contrast, Great Black-backed Gulls and Glaucous Gulls had detectable anti-H5 antibodies only in 2023, after the introduction of HPAIV.

Discussion

We investigated AIV seroprevalence in 916 seabirds and 157 gulls across multiple species and locations in Atlantic Canada before (2011–2016) and after (2022–2023) the arrival of HPAIV in the region in late 2021. Birds sampled in 2022 and 2023 (600 seabirds and 77 gulls) were also tested for active AIV infection. While active infections were detected only in a small number of individuals, the seroprevalence data indicate that most of the species included in this study were exposed to AIV. Notably, we found that anti-NP and anti-H5 seroprevalence varied among species and across years.

We acknowledge that the absence of formal statistical analyses is a limitation of this study. Given the uneven and often limited sample size across species, years and locations, statistical testing was not feasible and would have provided limited interpretive value. Nevertheless, the patterns presented here still offer valuable insights into potential trends and serve as a foundation for generating hypotheses and guiding future research.

Anti-NP antibodies were detected in all species sampled before the arrival of HPAIV (Figs. 2 and 3), while anti-H5 antibodies were only detected in Atlantic Puffins, Black-legged Kittiwakes, Razorbills, and sub-adult and adult gulls prior to the outbreak. Anti-H5 seroprevalence among seabirds and gulls sampled before the arrival of HPAIV is presumably due to past infection with LPAIV H5 viruses which are known to circulate among wild birds in the region (Huang et al. 2013) and not due to exposure during the 2014/2015 HPAIV outbreak that only affected birds in western and central North America (Pasick et al. 2015; Bevins et al. 2016; Ramey et al. 2016).

In 2022, the majority of HPAIV-associated wild bird deaths in eastern Canada were observed in Northern Gannets, Common Murres, and Common Eiders, while notable mortalities due to HPAI were also observed in gulls, cormorants, Atlantic Puffins, Black-legged Kittiwakes, Razorbills, and terns (Avery-Gomm et al. 2024). In the current study, the highest prevalence of both active infections and seropositivity was in Common Murres in 2022. This is consistent with Common Murres being one of the seabird species most impacted during the 2022 outbreak in Atlantic Canada (Avery-Gomm et al. 2024). The Common Murres sampled in this study after the introduction of HPAI were sampled from Gull Island, the same colony at which Black-legged Kittiwakes and Atlantic Puffins were also sampled. Based on the timing of detection of antibodies in these individuals, the outbreak at this colony likely began in mid- to late-July 2022. Although this H5N1 virus (genotype A1) (Signore et al. 2025) spread through multiple species at the Gull Island colony around the same time, active infections, and seroprevalence varied between species.

This can likely be attributed to differences in pre-existing immunity, nesting habitats and densities, interaction rates with con- and allospecifics and breeding and foraging behaviours (Ainley et al. 2021; McPhail et al. 2024).

After the arrival of HPAIV in 2022, significant mortalities were reported at some colonies while no mass die-offs were observed at others, such as Gull Island (Avery-Gomm et al. 2024; McLaughlin et al. 2024) or Kent Island where HPAI mortalities were detected during the nesting season (Taylor et al. 2023). Due to high breeding densities at Gull Island, it is difficult to monitor the actual number of breeding pairs on an annual basis, therefore it is possible that the populations at Gull Island may have also experienced significant mortality. For instance, a study in the UK showed that nonbreeding Common Murres quickly replaced breeding birds that failed to return the following year, thereby masking the true impact of the outbreak at a colony level (Birkhead and Hatchwell 2025). Conversely, given that anti-NP antibodies were detected in Atlantic Puffins, Black-legged Kittiwakes, and Common Murres on Gull Island before the introduction of HPAIV, pre-existing immunity may have played a crucial role in limiting seabird mortalities at this colony. Based on studies published between 1963 and 2014, AIV prevalence (active infections) globally in Common Murres was 3.5% (81/2330), with a seroprevalence of 33% (84/257) (Lang et al. 2016) while Wille et al. (2014) reported that the AIV prevalence rate among Common Murres in Newfoundland and Labrador between 2008 and 2011 was 5.4% (71/1317). During 2011, anti-NP seroprevalence was high among Common Murres in the region, at 44% and 45% at Gull Island and Cabot Island, respectively. At a single sampling event at Gull Island in 2011, 60 of 68 Common Murres tested positive for AIV RNA, with nearly all of the subtyped viruses being H1N2, indicating a likely localized outbreak of this strain (Wille et al. 2014). These previous studies support that at least two distantly located breeding populations of Common Murres were exposed to AIVs before the arrival of HPAIV. This may help to explain why, despite high seroprevalence detected in Common Murres in 2023 (Fig. 2), there were no significant mortality events for this species in 2023 (Cormier et al. 2024). The differential impact of HPAIV on Common Murre populations across colonies and years reflects differences in virus dynamics (Alkie et al. 2023; Signore et al. 2025) and colony-level susceptibility (McLaughlin et al. 2024), and pre-existing immunity (Wille et al. 2014). Results from the current study indicate that the Atlantic Puffin and Black-legged Kittiwake populations at Gull Island were also exposed to AIVs prior to the introduction of HPAI. This may have contributed to the high survival rates among adult puffins as population surveys conducted in 2023 and 2024 also show a positive population trend (Iles et al. 2025). In contrast, a population survey of Black-legged Kittiwakes around Gull Island in 2023 reported the lowest counts on record since annual surveys began in 2001 (Environment and Climate Change Canada 2023), indicating that kittiwakes may have been significantly impacted by HPAIV despite prior AIV exposure.

The 2022 HPAIV outbreak led to unprecedented mortality of Northern Gannets in Europe and North America, reducing breeding populations by more than 50% (Avery-Gomm et al. 2024; Lane et al. 2024; McPhail et al. 2024). Several factors likely contributed to this heightened susceptibility, including large, densely packed colonies and high levels of intra- and inter-specific interactions at sea (McPhail et al. 2024). Furthermore, Northern Gannets were found to be minimally exposed to AIVs prior to the widespread circulation of H5Nx viruses in Europe and the Americas that started in late 2020 (Wille et al. 2014; Lang et al. 2016). Based on studies published between 1963 and 2014 globally, active infections were detected in only 1 of 202 Northern Gannets sampled, including 76 sampled gannets in Newfoundland and Labrador between 2008 and 2011 (Wille et al. 2014; Lang et al. 2016). In the present study, samples from Northern Gannets were collected from Cape St. Mary's in 2022, prior to the arrival of HPAIV at this colony (Fig. 2). The lack of detectable antibodies in the sampled individuals provides a possible explanation as to why Northern Gannets were so severely impacted by HPAI. These birds may not have had the same level of pre-existing immunity as other species, presumably due to lack of previous exposure to LPAIVs. However, this may be confounded as most samples from 2022 (11/14) were collected from juveniles that were seronegative. The seroprevalence of anti-NP and anti-H5 antibodies in 2023 indicates that survivors in this population did develop some level of immunity following the arrival of HPAIV in 2022. While other studies have reported recovery of Northern Gannets following infection with HPAIV (Grémillet et al. 2023; Lane et al. 2024), the durability of the immune response and the effectiveness in preventing future infection or protecting against fatal outcomes remains unknown.

No active infections were detected in Leach's Storm Petrels despite these birds breeding in large colonies with millions of individuals (Pollet et al. 2021). Seroprevalence of anti-NP antibodies was very low (6%) and no anti-H5 antibodies were detected ($n = 118$). Moreover, mortality rates of Leach's Storm-Petrel remained at normal levels during the 2022 outbreak, indicating that HPAI likely did not cause mass mortality in this species. Based on studies published between 1963 and 2014 globally, AIV prevalence has been low among Leach's Storm Petrels (0.94%, 12/1272) (Lang et al. 2016), with no active infections in this species in Newfoundland and Labrador between 2008 and 2011 (Wille et al. 2014). Leach's Storm Petrels are nocturnal, nest in burrows, and rarely interact with allospecifics during foraging due to differences in range (Hedd et al. 2018; Ronconi et al. 2022) and as such are presumably less likely to become infected with AIVs.

Our study revealed distinct differences in seroprevalence between juvenile and sub-adult and adult gulls (Fig. 3). All juveniles were seronegative across all sampled years, indicating no previous exposure to AIV. This aligns with findings from other studies that highlight the differences in viral exposure based on age, with younger gulls typically having lower exposure rates than adults due to their limited movements and interactions with other species (Farnsworth et al. 2012; Spivey et al. 2017).

Given the short window of viral RNA detection during AIV active infections (Kim et al. 2014; Wight et al. 2024), serological studies are vitally important to understand and document past exposures to AIV. The presence of antibodies indicates prior infection or exposure even when active infections are

no longer detectable by real-time RT-PCR, allowing gaps in surveillance to be filled. The immune response to AIVs in wild birds varies depending on virus strain (Tarasiuk et al. 2023), age of the bird (Farnsworth et al. 2012; Spivey et al. 2017; Ineson et al. 2022), and past AIV infections (Fereidouni et al. 2009; Berhane et al. 2010; Costa et al. 2011). Past AIV exposure may confer some level of heterosubtypic immunity (Koethe et al. 2020; Tarasiuk et al. 2023), potentially reducing susceptibility to HPAIV or limiting severe disease. However, the degree of protection against HPAIVs likely varies substantially and reinfections can occur with waning immunity (Caliendo et al. 2022*a*; Wight et al. 2024).

Conclusions

Our findings demonstrate the importance of longitudinal serological studies to understand AIV dynamics in wild birds. While AIVs have historically circulated in several seabird and gull populations in Atlantic Canada at low levels, the arrival of HPAI in late 2021 altered AIV dynamics in many of these populations. We found that seroprevalence was often similar within the same species and year even at different locations, and that among gulls, seroprevalence was lower in juvenile birds compared to sub-adults and adults. Pre-existing immunity to AIVs prior to the arrival of HPAI in Atlantic Canada varied by both species and location. This provides a likely explanation for why certain species and colonies were more severely impacted and experienced higher mortality rates due to HPAIV. Serological studies provide valuable insights about species and/or locations that lack pre-existing immunity to AIVs and as such may help identify those most likely to be negatively impacted by future outbreaks. Overall, this study emphasizes the vital importance of continued surveillance and serological studies to better understand the evolving ecology of AIV in seabirds, not only within North America, but globally.

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Data availability

The metadata and the screening results for all the birds included in this study have been included in the Supplementary Data file.

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Competing interests

The authors declare no conflict of interest.

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Supplementary material

Supplementary data are available with the article at https://doi.org/10.1139/cjm-2025-0195.

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